

Manuscript

Layer-1

稿件草稿 HTML 版（基于 docs/manuscript_draft.md）

Self-contained academic HTML with embedded SciencePlots figures (base64).

稿件草稿 / Manuscript draft (Layer-1 exploratory surrogate)

EN title: Literature-anchored low-order surrogate for exploratory screening of indirect mitral annuloplasty strategies

中文工作题: 文献锚定的低阶代理模型：间接二尖瓣成形策略的探索性筛查

工作短题: FMR IMA Layer-1 exploratory surrogate

Venue class: methods / computational biomechanics software paper (CMBBE / MedEngPhys / RSOS-class) — not a clinical decision trial.

One-sentence argument: We provide a transparent, literature-anchored algebraic surrogate that reproduces selected Galili peak-systole anchors and ranks exploratory IMA-CS / IMA-AP scenarios under explicit geometric and screening assumptions—without claiming new LHHM FEA, clinical recommendation, or independent external validation of blended cases.

Data: `python run_pipeline.py --seed 42 --paper` (reproducible Layer-1 outputs only).

Verification levels: Level 0 = calibrated to / reproduced selected Galili cases; Level 1 = exploratory planning map + scenario ranker. Level 2 patient-specific LHHM/FSI is out of scope.

Abstract

Background: Computational IMA studies (Galili et al., *R. Soc. Open Sci.* 2022) report suture/bridge shortening and peak-systolic geometry/leakage. Those percentages are mechanical parameterizations. Peak-systolic IMA-AP 50% yields AP **15.9 mm** (near-direct AP effect), not undeformed diastole **34.4 mm**. Clinical device classes also differ: MAVERIC evaluates the **ARTO** (CS–IAS / IMA-AP-class) system; Carillon / REDUCE-FMR are IMA-CS-class.

Methods: We implement a reproducible Python Layer-1 surrogate: an algebraic mechanics proxy plus a literature-calibrated leakage proxy (SPH-inspired). Two maps are available—(i) Galili peak-systole geometry reproduction; (ii) an exploratory planning map using assumption-prior transfer efficiencies η (not clinically calibrated constants; η_{CS} is **not** fitted from MAVERIC/ARTO). A discrete scenario ranker minimizes physics leakage-proxy regurgitation subject to an AP ceiling, an

illustrative NiTi engineering screen, and a Rottländer CS–LCx **risk-screening** threshold. Dual-suture commissural $\times 0.5$ is an explicit hypothesis parameter. Main figures use physics only; YAML anchor blending is restricted to calibration case IDs and constitutes **reproduction**, not external validation.

Results (seed=42): Under the planning map the ranker evaluates 36 points and retains 30 feasible designs. Best candidate under stated assumptions: **IMA-AP dual suture 60%** (assumption $\eta_{ap}=0.30 \rightarrow$ AP reduction **18.0%**, physics leakage-proxy $\sim 0.074\%$, jet=central). Best feasible IMA-CS on default anatomy: bridge **20%** (CS–LCx **8.6 mm** at the risk-screen edge). Galili peak-systole map: IMA-AP 50% AP **15.9 mm**; IMA-AP 70% AP **12.4 mm** with commissural leak and higher leakage than 50%.

Conclusions: A literature-anchored low-order surrogate can support **exploratory screening** of IMA strategy settings with inspectable assumptions. It is not a preoperative clinical decision system, does not equate physics % with clinical regurgitant volume, and must not treat high anchor-weight reproduction as independent validation.

Keywords: functional mitral regurgitation; indirect mitral annuloplasty; surrogate model; exploratory screening; coronary sinus; Layer-1

中文摘要（要点）

文献锚定的低阶代理，用于 IMA-CS / IMA-AP 探索性筛查；纠正 Galili 峰缩期 AP 映射与 MAVERIC=ARTO 归类； η 为假设先验；输出为假设下最优候选而非临床推荐；physics % $\neq$ 临床反流容积。

1. Introduction

Functional mitral regurgitation (FMR) reflects ventricular remodeling, annular dilatation, and impaired leaflet coaptation. Indirect mitral annuloplasty (IMA) concepts remodel annular geometry without direct leaflet repair. **IMA-CS** (Carillon-class coronary-sinus devices) and **IMA-AP** (CS–IAS suture; ARTO / MAVERIC mechanism class) are not mechanically interchangeable.

Galili et al. (2022) compared generic IMA-CS and IMA-AP in the Living Heart Human Model and reported peak-systolic AP, ROA, and leakage. IMA-AP acts nearly directly on AP diameter (disease 26.1 mm $\rightarrow$ IMA-AP 50% $\rightarrow$ **15.9 mm**). Undeformed diastolic AP 34.4 mm is a different cardiac phase and must not be paired with peak-systolic ROA/leakage without phase tags.

What remains for software methods is transparent **dose translation and exploratory ranking** under stated assumptions—not a claim of first CS-vs-AP comparison, not new FEA, and not a clinical

recommendation engine. MAVERIC AP pairs (~14–15%) contextualize ARTO / IMA-AP-class geometry; Carillon TITAN II (~15% AP) and REDUCE-FMR (regurg ↓ directionality) contextualize IMA-CS without fabricating η_{CS} from MAVERIC.

2. Methods

Pathology and geometry. Parametric papillary pathology (44% posterior passive). Undeformed diastole: AP 34.4 mm, annulus 118.5 mm. Galili peak-systole disease AP 26.1 mm.

Device kinematics. `models/devices.py`: Galili mode interpolates published peak-systolic AP; clinical/planning mode uses assumption-prior η or preferred `target_ap_reduction_pct`. $\eta_{ap}=0.30$ and $\eta_{cs}=0.55$ are **assumption priors** (`configs/design_space.yaml`). Invalid prior $\eta_{CS}\approx 0.668$ from MAVERIC→Carillon is removed.

Mechanics and leakage proxies. `run_mechanics_proxy` (alias `run_fea_surrogate`) returns uncalibrated coaptation-gap / strain / contact proxy scores. `sph/hemodynamics.py` is a literature-calibrated leakage proxy. Physics regurg % $\neq$ clinical regurgitant volume.

Calibration vs held-out. Calibration IDs: `pathology`, `ima_cs_22`, `ima_ap_50`. Held-out: `ima_cs_14/18`, `ima_ap_30/70` with leave-one-out plan (`configs/surrogate_calibration.yaml`). High blend weight = reproduction.

Scenario ranker. Minimizes physics leakage proxy subject to $AP \leq 20\%$, NiTi alternating strain $\leq 0.4\%$ (engineering screen), $CS-LCx \geq 8.6$ mm (Rottländer risk screen; prefer patient-measured $CS-LCx$; cinch slope $11-0.12\times$ shortening is an assumption). JSON: `best_candidate` (+ `compat` recommended).

Dual suture. Commissural penalty $\times 0.5$ is a hypothesis parameter; results are hypothesis-generating.

3. Results (seed=42)

3.1 Galili peak-systole anchors (reproduction)

Case	Leakage %	Peak-sys AP mm	ROA mm ²
pathology	5.26	26.1	172.8
ima_cs_22	0.29	24.8	48.2

ima_ap_50	0.08	15.9	27.3
ima_ap_70	0.13	12.4	46.1

Surrogate blended reports at calibration IDs reproduce leakage within test tolerances. Intermediate / held-out leakage is physics-driven and not claimed as validated.

3.2 Exploratory scenario ranking (planning map)

- Evaluated **36** / feasible **30**
- **Best candidate under assumptions:** IMA-AP dual **60%**, AP **18.0%**, physics **~0.074%**, jet=central, commissural fraction **0.165**
- Single-suture 60%: physics **~0.075%**, jet=mixed, commissural fraction **0.33**
- Best IMA-CS: **20%** bridge, AP **11.0%**, physics **~0.156%**, CS–LCx **8.6 mm**, NiTi alt. strain **0.34%**

$\eta \pm 20\%$ (assumption-prior sensitivity): dual 70% (η^- , AP 16.8%, physics $\sim 0.044\%$); dual 50% (η^+ , AP 18.0%, physics $\sim 0.112\%$). Not FEA UQ.

3.3 Directionality-only clinical context

ARTO/MAVERIC: AP↓. Carillon TITAN II: $\sim 15\%$ AP context. REDUCE-FMR: regurg↓. Magnitudes not equated to physics %.

4. Discussion / limitations

Central advance: transparent literature-anchored exploratory screening software with corrected Galili peak-systole AP semantics and correct MAVERIC=ARTO device-class mapping. Limitations: algebraic proxies; held-out Galili leakage (esp. AP70) not tightly reproduced without blend; no patient-specific validation; η are priors; LCx/NiTi screens are literature/engineering screens not safety clearances.

5. Conclusion

A Layer-1 literature-anchored surrogate can rank exploratory IMA scenarios under explicit assumptions after correcting peak-systole Galili AP facts and MAVERIC device-class attribution. It

should be cited as methods/software for exploratory screening—not as clinical preoperative decision support.

Collab / Forbidden-claim notes: see docs/chatgpt_collab/20260913_p0_review_response.md and docs/paper_framework_nature.md (scaffold, not main-text claims).

Embedded SciencePlots figures

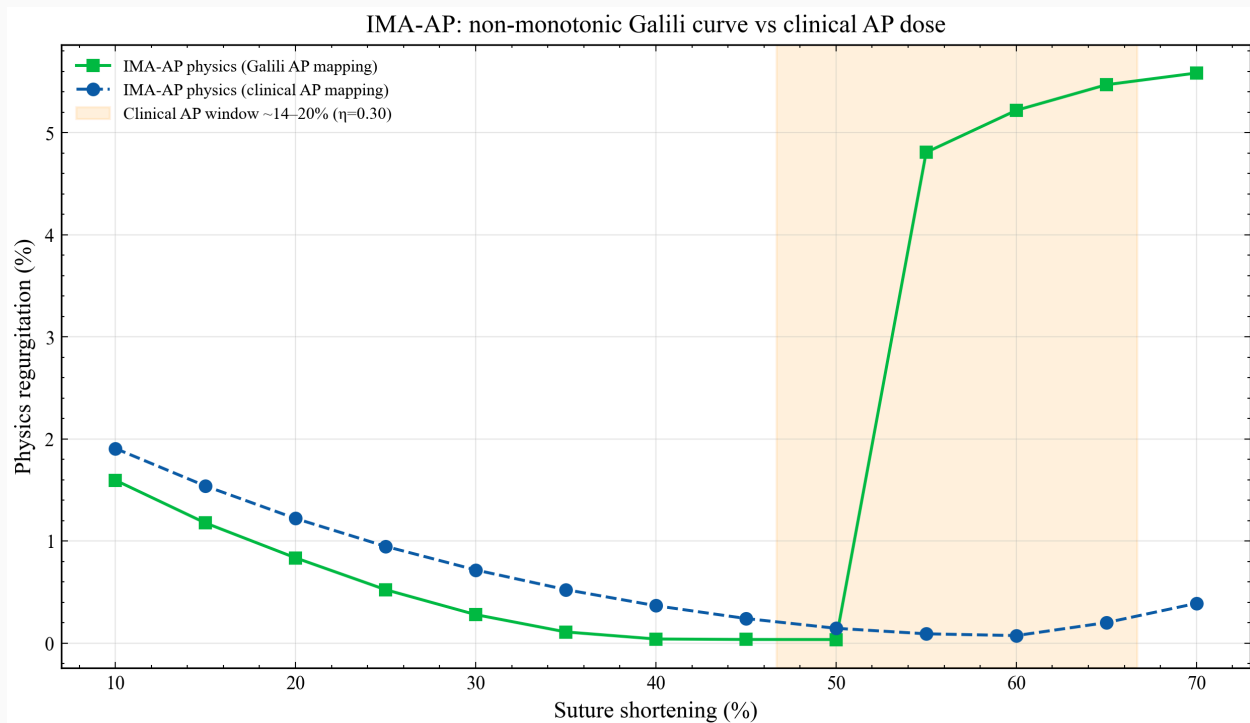


图 1. IMA-AP 物理反流随缝线缩短百分比的变化：Galili 映射 vs 临床映射

文件：fig1_ima_ap_nonmonotonic_clinical_window.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：本图回答“计算缝线缩短百分比能否直接当作临床前后径（AP）剂量？”
 横轴为 IMA-AP 缝线缩短 %；
 纵轴为 Layer-1 泄漏代理 physics_regurgitation_pct（代数力学代理 + 文献校准泄漏代理；不是超声反流容积分数）。

如何读：（1）Galili 映射复现发表峰缩期 AP：IMA-AP 50% → 峰缩期 AP=15.9 mm（相对疾病 26.1 mm 约 39% AP 缩减），
 并非未变形舒张期 34.4 mm / 0% AP；70% → 12.4 mm 并交界区泄漏升高。（2）规划映射采用假设先验
 $AP_reduction\% = \eta \times shortening\%$ ，默认 $\eta_{ap}=0.30$ （假设，非临床标定），使 50% 缝线对应约 15% AP，落入
 ARTO/MAVERIC（IMA-AP 类，不是 Carillon）约 14–15% 对话窗口。绿色带为约 14–20% AP 窗口。

结论（仅限本代理）：规划映射下窗口内总体可筛查；Galili 峰缩期映射下 70% 非单调恶化。
 勿将纵轴等同临床试验反流容积。

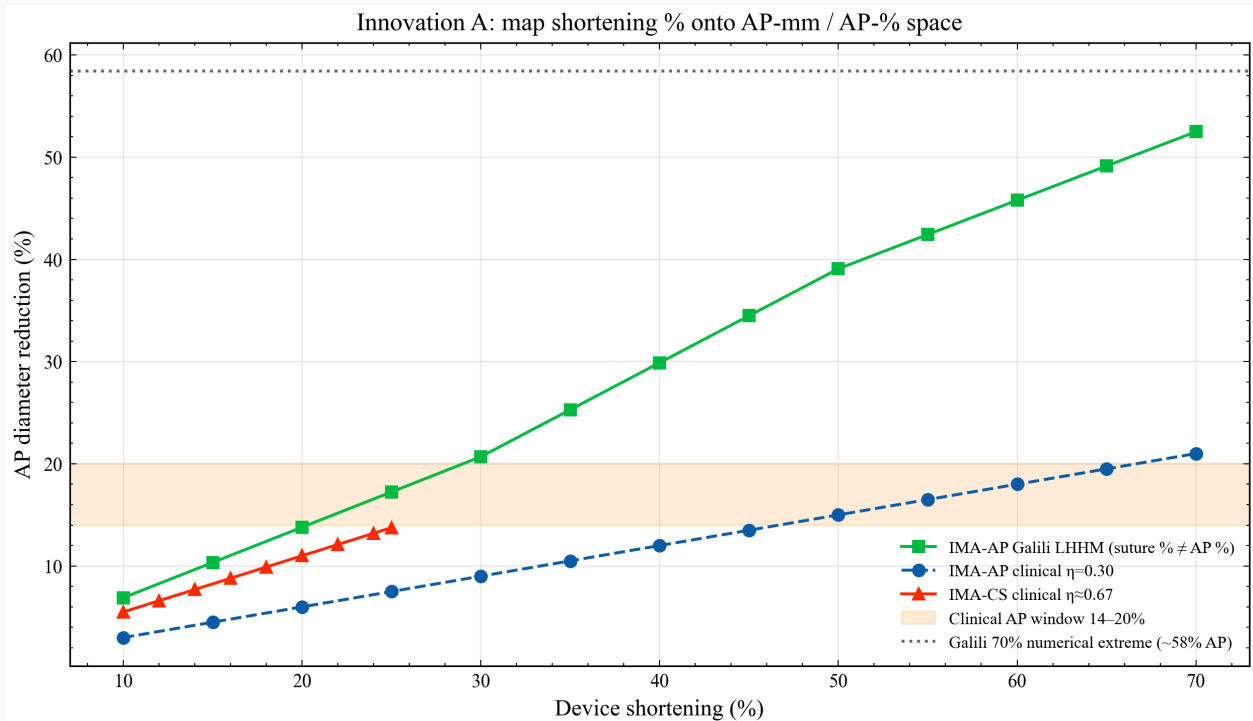


图 2. 装置缩短百分比到 AP 直径缩减百分比的剂量映射

文件: fig2_suture_vs_ap_reduction.png (SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明)

来龙去脉与读图说明: 横轴为缩短 %; 纵轴为 AP 缩减 %。轨迹对照: Galili 峰缩期 (IMA-AP 近直接缩 AP)、

规划 IMA-AP (假设 $\eta_{ap}=0.30$)、规划 IMA-CS (假设 $\eta_{cs}=0.55$, 非 MAVERIC 拟合; Carillon/TITAN II ~15% 仅语境)。

MAVERIC=ARTO (IMA-AP 类)。水平带 14–20% 为探索性规划窗口。

如何读: 同一缩短 % 在不同映射下 AP 缩减截然不同。 η 为假设先验, 不是成像–FEA 辨识, 也不是“临床标定常数”。

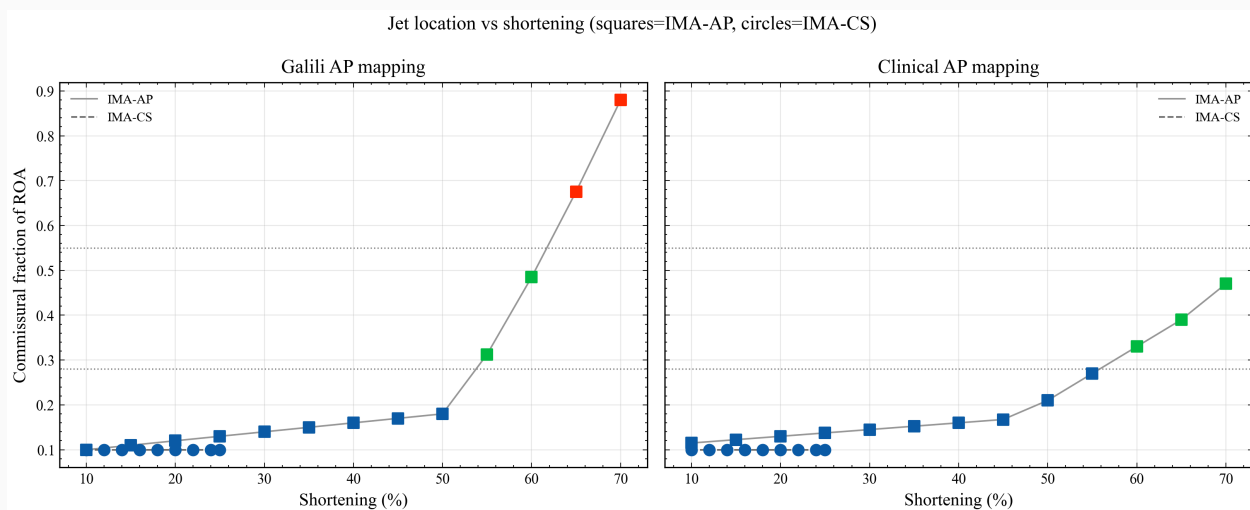


图 3. 射流位置与交界区 ROA 分数随缩短变化（Galili vs 临床）

文件: fig3_jet_location.png（SciencePlots + Times New Roman, dpi≥300; 坐标轴为英文标签, 释义见下文中文说明）

来龙去脉与读图说明：反流“量”之外，规划还关心泄漏机制位置。本代理将 ROA（regurgitant orifice area,

反流口面积，单位 mm^2 ，由对合间隙等代理量估计）拆为中央份额与交界区份额，并分类为 central /

commissural / mixed。横轴为缩短 %；纵轴为交界区 ROA 分数；方块=IMA-AP，圆点=IMA-CS；左右面板对照

Galili 与临床映射。水平虚线为分类阈值示意。

如何读：交界区分数升高意味着泄漏更偏交界区；分类从 central 变为 mixed/commissural 表示机制标签切换。

该标签是代理口上的机制草图，不是超声 PISA（proximal isovelocity surface area）或多普勒诊断。

seed-42 推荐双缝线 60% 的机制来源可在本图与图 5 对照中追溯：同 AP 下交界份额更低、jet 保持 central。

结论：在临床映射下，IMA-AP 高剂量单缝线更易出现 mixed；IMA-CS 在可行桥缩短范围内多保持 central。

结合图 5，双缝线可在匹配 AP 缩减下压低交界区分数。

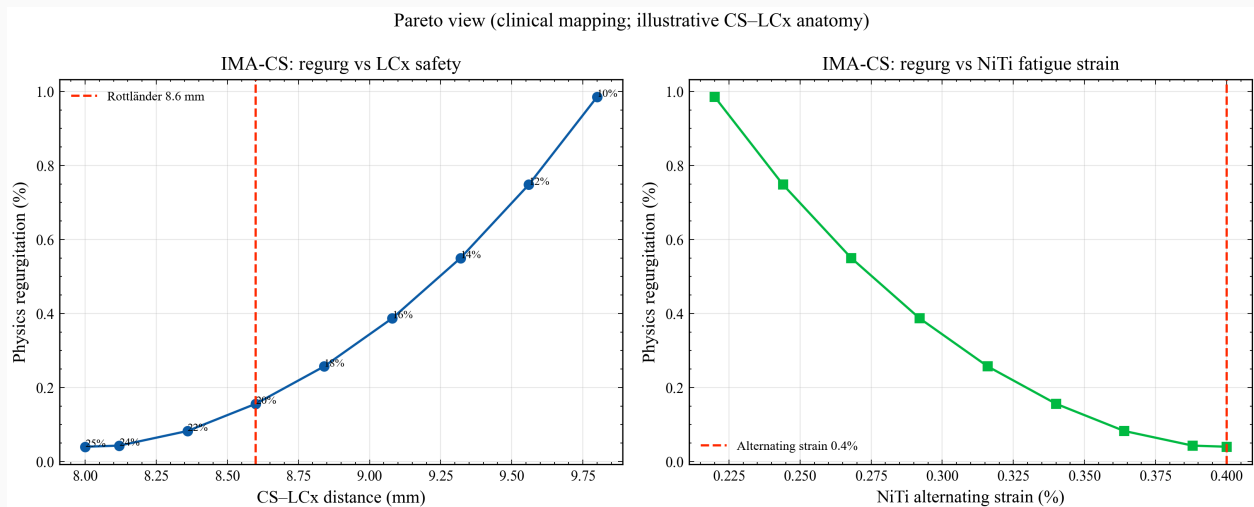


图 4. IMA-CS 临床映射：反流-筛查 Pareto（CS-LCx 与 NiTi 交变应变）

文件：fig4_pareto_lcx_strain.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：IMA-CS 在降低反流的同时可能压缩 CS-LCx 距离（coronary sinus-left circumflex artery，

冠状窦与左回旋支间距）。文献筛查边界取远端着陆区 CS-LCx ≥ 8.6 mm（Rottländer 等；注意：<8.6 mm

预测妥协 ≠ ≥8.6 mm 已证明安全）。同时对 NiTi（镍钛合金桥）交变应变设置 <0.4% 的工程筛查上限。

默认示意解剖基线 CS-LCx=11.0 mm，可替换为患者 CT。标题用“筛查”而非“安全证明”。

如何读左右面板：横轴多为筛查指标（CS-LCx 或交变应变），纵轴为 physics 反流；点随桥缩短移动，

越接近约束边界，可行域越窄。22% 及以上在默认解剖上常因 CS-LCx 违约束而不可行。

结论：默认解剖上可行最优贴近 CS-LCx=8.6 mm 的桥缩短 20%。该结果是规划器筛查信号，不是患者级安全证明。

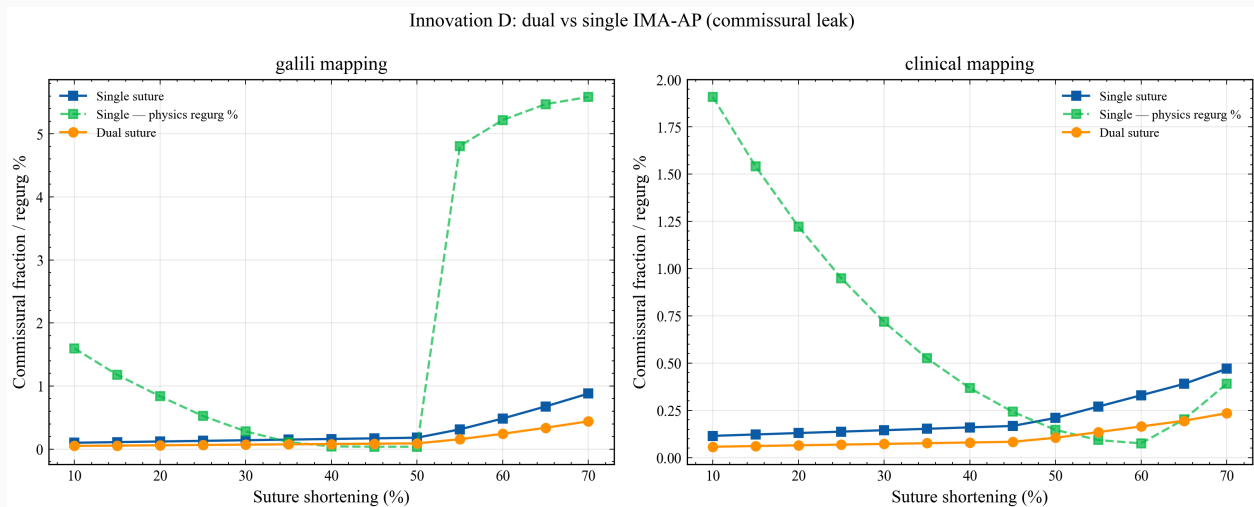


图 5. 双缝线 vs 单缝线 IMA-AP：匹配 AP 缩减下的交界区泄漏对照

文件：fig5_dual_vs_single_suture.png（SciencePlots + Times New Roman，dpi≥300；坐标轴为英文标签，释义见下文中文说明）

来龙去脉与读图说明：可选贡献 C4（Innovation D）在相同缝线缩短 %（故相同临床 AP 缩减）下比较单/双缝线。

纵轴关注交界区 ROA 分数；虚线可叠加单缝线 physics 反流作对照。左右面板对照 Galili / 临床映射。

本图隔离“缝线数量”变量，回答机制问题，不是器械清关或产品声明。

如何读：在 60% 临床映射点，单缝线 jet=mixed、交界分数 0.330；双缝线 jet=central、交界分数 0.165，

physics 反流由约 0.160% 降至约 0.152%——分类改善更醒目，幅度改善相对温和。70% 点 AP 缩减 21%

超出默认 20% 规划上限，仅作机制对照，不作推荐植入。

结论：双缝线是机制草图；seed-42 场景排序最优候选双缝线 60% 正是在该对照下选出的网格点。

勿将 0.152% physics 等同于临床反流容积改善。